

Project Collaboration & Contribution Report
Project Title: Automated Ribosomal Analysis
Date: May 4, 2026

Executive Summary of Contributions

While this project was intended to be a collaborative effort, the final output was primarily the result of individual initiative. I was responsible for the end-to-end development of the project, including the conceptual framework, the technical implementation (coding), and the primary drafting of the technical report. My partner's contributions were limited to secondary grammatical proofreading of the final document.

Technical & Research Development

I independently managed the core pillars of the project:

Conceptualization: Defined the project scope regarding ribosomal protein analysis and established the research methodology.

Software Architecture: Developed the entire codebase for the automated analysis. Due to technical hurdles, my partner was unable to access or review the local development environment or the source code throughout the duration of the project.

Data Integration: Managed the "base-stacking" and "interface" datasets, ensuring the synthesis of data structures with structural biology requirements.

Workload Distribution

Primary Researcher/Developer (Self): 95%

Literature review, algorithm design, coding, data analysis, and technical writing.

Project Partner: 5%

Correction of syntax and grammatical errors in the final written report.

Collaboration Challenges

Throughout the project lifecycle, there was a significant gap in conceptual understanding. My partner did not demonstrate a foundational grasp of the project's objectives or the underlying biological mechanisms being modeled. Despite attempts to integrate their efforts, the collaboration remained one-sided, resulting in my partner serving as a minimal editor for the final draft rather than a technical contributor.